

A method to correct the optical flame transfer functions in acoustically forced, partially premixed hydrogen flame

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1. Introduction

Thermoacoustic instabilities arise when fluctuations in heat release couple with acoustic waves in the combustor, forming a feedback loop that can amplify pressure oscillations and lead to severe damage or even catastrophic failure. The response of the flame's overall unsteady heat release rate (HRR) to acoustic or equivalence ratio perturbations can be characterized by flame transfer function (FTF). Accurate experimental characterization of HRR is essential, yet direct measurements remain impractical because they require simultaneous determination of multiple species and temperatures.

In practice, chemiluminescence intensity (e.g., CH* and OH* in hydrocarbon fuels) is a widely used indicator of HRR in fully premixed flames. However, this approach becomes less reliable under partially premixed conditions, where equivalence ratio fluctuations strongly influence emission intensity. In this context, the FTF obtained through chemiluminescence measurements will be referred to as the optical FTF. Developing reliable methods to construct FTFs under these conditions remains a pressing challenge.

Motivated by the challenges discussed above, this study evaluates a correction strategy for the optical FTF, using chemiluminescence intensity and flame surface area to construct FTFs in an acoustically forced, partially premixed hydrogen flame.

2. Methodologies

The experiments were conducted using the same setup as reported in our previous work [1]. All measurements were performed at ambient temperature and atmospheric pressure, with a global equivalence ratio $\phi_0 = 0.65$ and a mean bulk velocity $\bar{u}_0 = 18$ m/s. The perturbation intensity was fixed at 5% of the mean velocity, and the forcing frequency was varied from 60 to 600 Hz with an increment of 20 Hz.

For flame front visualization, Al₂O₃ particle tracers ($\sim 0.1 \mu\text{m}$) were seeded for Mie scattering. Taking the flame thickness (~ 0.4 mm, calculated using Cantera) as the flow characteristic length scale, the corresponding Stokes number is below 0.01, indicating that the tracer particles can reliably resolve small-scale flame structures. As shown in Fig.1(a), a 532 nm laser sheet was aligned with the central axial plane, and a high-speed camera (Phantom VEO 1310L) operating at 10 kHz in single-frame mode was used to record the transient evolution of the flame front. Simultaneously, OH* chemiluminescence was measured in synchronization to obtain optical FTFs. The transient flame front detection, as shown in Fig.1(b), was using a hybrid algorithm [2], which combined Otsu segmentation and Canny edge detection.

Assuming small perturbations, the transfer function F_I , relating the global OH* intensity fluctuations (I') to the velocity fluctuations at the reference section (u'), is corrected using the flame surface area transfer function F_A , in conjunction with quasi-steady relationships linking the equivalence ratio (ϕ) to the laminar flame speed (s_L), heat of reaction (h_R), and OH* intensity.

3. Results and discussions

Fig. 1(d) compares the directly measured F_I with the corrected flame transfer function F . Here, F_A is reconstructed from the extracted two-dimensional flame front under the assumption of axisymmetry. Overall, F deviates noticeably from F_I , particularly in the low-frequency range where equivalence ratio fluctuations are significant, while the discrepancies gradually diminish at higher frequencies.

It should be noted that the present results have not yet been validated against the true flame transfer function, and the axisymmetric assumption may introduce some uncertainty. Further quantitative validation against reference measurements will be pursued in future work. Nevertheless, the proposed methodology has been previously validated for laminar flames under equivalence ratio modulation, providing partial support for its applicability, although those results are not shown here. The present work is aiming to extend this to a turbulent flame.

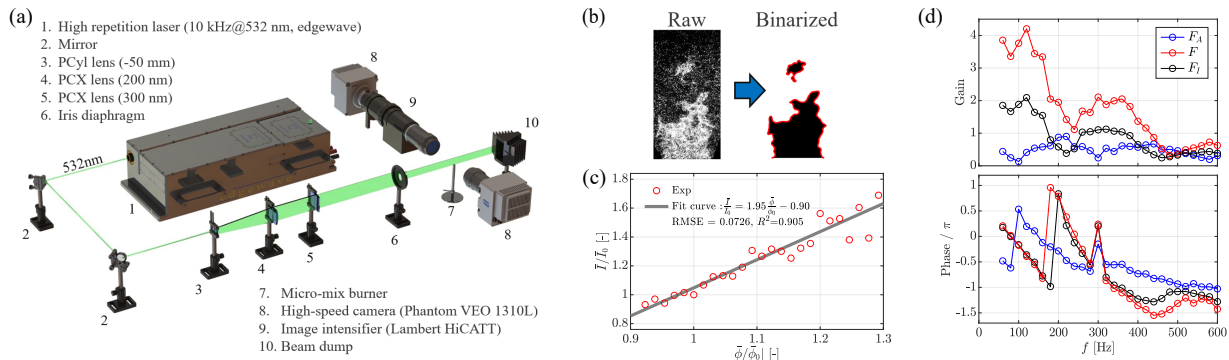


Fig. 1. (a) Experimental setup; (b) Flame front detection; (c) Steady-state relationship between time-averaged global OH* intensity and global equivalence ratio; (d) Directly measured and corrected FTFs (F_I and F), along with the corresponding flame surface area transfer function (F_A).

References

- [1] W. Liang, Z. Zhao, C. Miao, B. Sa, B. Zhou, D. Yang, Energy (2026) 140827.
- [2] O. Chaib, Y. Zheng, S. Hochgreb, I. Boxx, Experiments in Fluids 64 (5) (2023) 104.